

Solving Quadratic Equations by Completing the Square (Part II)

Why do we divide the coefficient of x by two and square the result to complete the square?

Expand a generic perfect square.

$$(x+a)^2 = (x+a)(x+a) = x^2 + xa + ax + a^2 = x^2 + 2ax + a^2$$

We know the result is a perfect square.

This gives us a relation between the coefficient of x and the number we need to complete the square.

$$x^2 + 2ax + a^2 = (x+a)(x+a) = (x+a)^2$$

I) Divide the coefficient by two: $\frac{2a}{2} = a$

II) Then square the result: $a \cdot a = a^2$

This is the number we need to complete the square.

Solve by completing the square.

$$\textcircled{1} x^2 + 9x + 18 = 0$$

$$x^2 + 9x = -18$$

$$x^2 + 9x + \frac{81}{4} = -18 + \frac{81}{4}$$

$$(x + \frac{9}{2})(x + \frac{9}{2}) = -\frac{18}{1} + \frac{81}{4}$$

$$(x + \frac{9}{2})^2 = -\frac{72}{4} + \frac{81}{4}$$

$$(x + \frac{9}{2})^2 = \frac{9}{4}$$

$$\sqrt{(x + \frac{9}{2})^2} = \pm \sqrt{\frac{9}{4}}$$

$$x + \frac{9}{2} = \pm \frac{\sqrt{9}}{\sqrt{4}}$$

$$x + \frac{9}{2} = \pm \frac{3}{2}$$

$$x = -\frac{9}{2} \pm \frac{3}{2}$$

$$x = -3 \text{ or } x = -6$$

$$\textcircled{2} 0 = x^2 - 5x - 50$$

$$50 = x^2 - 5x$$

$$\frac{25}{4} + 50 = x^2 - 5x + \frac{25}{4}$$

$$\frac{25}{4} + \frac{50}{1} = (x - \frac{5}{2})(x - \frac{5}{2})$$

$$\frac{25}{4} + \frac{200}{4} = (x - \frac{5}{2})^2$$

$$\frac{225}{4} = (x - \frac{5}{2})^2$$

$$\pm \sqrt{\frac{225}{4}} = \sqrt{(x - \frac{5}{2})^2}$$

$$\pm \frac{\sqrt{225}}{\sqrt{4}} = x - \frac{5}{2}$$

$$\pm \frac{15}{2} = x - \frac{5}{2}$$

$$\frac{5}{2} \pm \frac{15}{2} = x$$

$$x = 10 \text{ or } x = -5$$

$$\textcircled{3} x^2 - 7x + 12 = 0$$

$$x^2 - 7x = -12$$

$$x^2 - 7x + \frac{49}{4} = -12 + \frac{49}{4}$$

$$(x - \frac{7}{2})(x - \frac{7}{2}) = -\frac{12}{1} + \frac{49}{4}$$

$$(x - \frac{7}{2})^2 = -\frac{48}{4} + \frac{49}{4}$$

$$(x - \frac{7}{2})^2 = \frac{1}{4}$$

$$\sqrt{(x - \frac{7}{2})^2} = \pm \sqrt{\frac{1}{4}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{1}}{\sqrt{4}}$$

$$x - \frac{7}{2} = \pm \frac{1}{2}$$

$$x = \frac{7}{2} \pm \frac{1}{2}$$

$$\boxed{x = 4 \text{ or } x = 3}$$

$$\textcircled{4} 0 = x^2 + 9x + 8$$

$$\frac{81}{4} - 8 = x^2 + 9x + \frac{81}{4}$$

$$\frac{81}{4} - \frac{32}{1} = (x + \frac{9}{2})(x + \frac{9}{2})$$

$$\frac{81}{4} - \frac{32}{1} = (x + \frac{9}{2})^2$$

$$\frac{49}{4} = (x + \frac{9}{2})^2$$

$$\pm \sqrt{\frac{49}{4}} = \sqrt{(x + \frac{9}{2})^2}$$

$$\pm \frac{\sqrt{49}}{\sqrt{4}} = x + \frac{9}{2}$$

$$\pm \frac{7}{2} = x + \frac{9}{2}$$

$$-\frac{9}{2} \pm \frac{7}{2} = x$$

$$\boxed{x = -1 \text{ or } x = -8}$$

$$\textcircled{5} 0 = x^2 - 3x + 2$$

$$-2 = x^2 - 3x$$

$$\frac{9}{4} - 2 = x^2 - 3x + \frac{9}{4}$$

$$\frac{9}{4} - \frac{2}{1} = (x - \frac{3}{2})(x - \frac{3}{2})$$

$$\frac{9}{4} - \frac{8}{4} = (x - \frac{3}{2})^2$$

$$\frac{1}{4} = (x - \frac{3}{2})^2$$

$$\textcircled{6} x^2 + 4x + 1 = 0$$

$$x^2 + 4x = -1$$

$$x^2 + 4x + 4 = -1 + 4$$

$$(x + 2)(x + 2) = 3$$

$$(x + 2)^2 = 3$$

$$\sqrt{(x + 2)^2} = \pm \sqrt{3}$$

$$x + 2 = \pm \sqrt{3}$$

$$\boxed{x = -2 \pm \sqrt{3}}$$

$$\textcircled{7} 0 = x^2 + 6x - 2$$

$$2 = x^2 + 6x$$

$$9 + 2 = x^2 + 6x + 9$$

$$11 = (x + 3)(x + 3)$$

$$11 = (x + 3)^2$$

$$\pm \sqrt{11} = \sqrt{(x + 3)^2}$$

$$\pm \sqrt{11} = x + 3$$

$$\boxed{-3 \pm \sqrt{11} = x}$$

$$\pm \sqrt{\frac{1}{4}} = \sqrt{(x - \frac{3}{2})^2}$$

$$\pm \frac{\sqrt{1}}{\sqrt{4}} = x - \frac{3}{2}$$

$$\pm \frac{1}{2} = x - \frac{3}{2}$$

$$\frac{3}{2} \pm \frac{1}{2} = x$$

$$\boxed{x = 2 \text{ or } x = 1}$$

$$\textcircled{8} \quad x^2 + x - 3 = 0$$

$$x^2 + x = 3$$

$$x^2 + x + \frac{1}{4} = 3 + \frac{1}{4}$$

$$(x + \frac{1}{2})(x + \frac{1}{2}) = \frac{3}{1} + \frac{1}{4}$$

$$(x + \frac{1}{2})^2 = \frac{12}{4} + \frac{1}{4}$$

$$(x + \frac{1}{2})^2 = \frac{13}{4}$$

$$\sqrt{(x + \frac{1}{2})^2} = \pm \sqrt{\frac{13}{4}}$$

$$x + \frac{1}{2} = \pm \frac{\sqrt{13}}{\sqrt{4}}$$

$$x = -\frac{1}{2} \pm \frac{\sqrt{13}}{2}$$

$$\boxed{x = \frac{-1 \pm \sqrt{13}}{2}}$$

$$\textcircled{9} \quad 0 = x^2 - 5x - 8$$

$$8 = x^2 - 5x$$

$$\frac{25}{4} + 8 = x^2 - 5x + \frac{25}{4}$$

$$\frac{25}{4} + \frac{8}{1} = (x - \frac{5}{2})(x - \frac{5}{2})$$

$$\frac{25}{4} + \frac{32}{4} = (x - \frac{5}{2})^2$$

$$\frac{57}{4} = (x - \frac{5}{2})^2$$

$$\pm \sqrt{\frac{57}{4}} = \sqrt{(x - \frac{5}{2})^2}$$

$$\pm \frac{\sqrt{57}}{\sqrt{4}} = x - \frac{5}{2}$$

$$\pm \frac{\sqrt{57}}{2} = x - \frac{5}{2}$$

$$\frac{5}{2} \pm \frac{\sqrt{57}}{2} = x$$

$$\boxed{\frac{5 \pm \sqrt{57}}{2} = x}$$

$$\textcircled{10} \quad x^2 + 3x + 1 = 0$$

$$x^2 + 3x = -1$$

$$x^2 + 3x + \frac{9}{4} = -1 + \frac{9}{4}$$

$$(x + \frac{3}{2})(x + \frac{3}{2}) = -\frac{1}{1} + \frac{9}{4}$$

$$(x + \frac{3}{2})^2 = -\frac{4}{4} + \frac{9}{4}$$

$$(x + \frac{3}{2})^2 = \frac{5}{4}$$

$$\sqrt{(x + \frac{3}{2})^2} = \pm \sqrt{\frac{5}{4}}$$

$$x + \frac{3}{2} = \pm \frac{\sqrt{5}}{\sqrt{4}}$$

$$x + \frac{3}{2} = \pm \frac{\sqrt{5}}{2}$$

$$x = -\frac{3}{2} \pm \frac{\sqrt{5}}{2}$$

$$\boxed{x = \frac{-3 \pm \sqrt{5}}{2}}$$

$$\textcircled{11} \quad 0 = x^2 - 7x - 4$$

$$4 = x^2 - 7x$$

$$\frac{49}{4} + 4 = x^2 - 7x + \frac{49}{4}$$

$$\frac{49}{4} + \frac{16}{4} = (x - \frac{7}{2})(x - \frac{7}{2})$$

$$\frac{49}{4} + \frac{16}{4} = (x - \frac{7}{2})^2$$

$$\frac{65}{4} = (x - \frac{7}{2})^2$$

$$\pm \sqrt{\frac{65}{4}} = \sqrt{(x - \frac{7}{2})^2}$$

$$\pm \frac{\sqrt{65}}{\sqrt{4}} = x - \frac{7}{2}$$

$$\pm \frac{\sqrt{65}}{2} = x - \frac{7}{2}$$

$$\frac{7}{2} \pm \frac{\sqrt{65}}{2} = x$$

$$\boxed{\frac{7 \pm \sqrt{65}}{2} = x}$$

$$\textcircled{12} \quad 2x^2 + 4x - 3 = 0$$

$$\frac{2x^2 + 4x - 3}{2} = \frac{0}{2}$$

$$x^2 + 2x - \frac{3}{2} = 0$$

$$x^2 + 2x = \frac{3}{2}$$

$$x^2 + 2x + 1 = \frac{3}{2} + 1$$

$$(x+1)(x+1) = \frac{3}{2} + 1$$

$$(x+1)^2 = \frac{3}{2} + \frac{2}{2}$$

$$(x+1)^2 = \frac{5}{2}$$

$$\sqrt{(x+1)^2} = \pm \sqrt{\frac{5}{2}}$$

$$x+1 = \pm \frac{\sqrt{5}}{\sqrt{2}}$$

$$x+1 = \pm \frac{\sqrt{5} \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$$

$$x+1 = \pm \frac{\sqrt{10}}{2}$$

$$\boxed{x = -1 \pm \frac{\sqrt{10}}{2}}$$

$$\textcircled{13} \quad 0 = 4x^2 - 40x + 5$$

$$\frac{0}{4} = \frac{4x^2 - 40x + 5}{4}$$

$$0 = x^2 - 10x + \frac{5}{4}$$

$$-\frac{5}{4} = x^2 - 10x$$

$$25 - \frac{5}{4} = x^2 - 10x + 25$$

$$\frac{25}{1} - \frac{5}{4} = (x-5)(x-5)$$

$$\frac{100}{4} - \frac{5}{4} = (x-5)^2$$

$$\frac{95}{4} = (x-5)^2$$

$$\pm \sqrt{\frac{95}{4}} = \sqrt{(x-5)^2}$$

$$\pm \frac{\sqrt{95}}{\sqrt{4}} = x-5$$

$$\pm \frac{\sqrt{95}}{2} = x-5$$

$$\boxed{5 \pm \frac{\sqrt{95}}{2} = x}$$

$$\textcircled{14} \quad 0 = -3x^2 + 18x + 4$$

$$\frac{0}{-3} = \frac{-3x^2 + 18x + 4}{-3}$$

$$0 = x^2 - 6x - \frac{4}{3}$$

$$\frac{4}{3} = x^2 - 6x$$

$$9 + \frac{4}{3} = x^2 - 6x + 9$$

$$\frac{9}{1} + \frac{4}{3} = (x-3)(x-3)$$

$$\frac{27}{3} + \frac{4}{3} = (x-3)^2$$

$$\frac{31}{3} = (x-3)^2$$

$$\pm \sqrt{\frac{31}{3}} = \sqrt{(x-3)^2}$$

$$\pm \frac{\sqrt{31}}{\sqrt{3}} = x-3$$

$$\pm \frac{\sqrt{31} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = x-3$$

$$\pm \frac{\sqrt{93}}{3} = x-3$$

$$\boxed{3 \pm \frac{\sqrt{93}}{3} = x}$$

$$\textcircled{15} \quad 9x^2 + 36x + 2 = 0$$

$$\frac{9x^2 + 36x + 2}{9} = \frac{0}{9}$$

$$x^2 + 4x + \frac{2}{9} = 0$$

$$x^2 + 4x = -\frac{2}{9}$$

$$x^2 + 4x + 4 = -\frac{2}{9} + 4$$

$$(x+2)(x+2) = -\frac{2}{9} + \frac{4}{1}$$

$$(x+2)^2 = -\frac{2}{9} + \frac{36}{9}$$

$$(x+2)^2 = \frac{34}{9}$$

$$\sqrt{(x+2)^2} = \pm \sqrt{\frac{34}{9}}$$

$$x+2 = \pm \frac{\sqrt{34}}{\sqrt{9}}$$

$$x+2 = \pm \frac{\sqrt{34}}{3}$$

$$\boxed{x = -2 \pm \frac{\sqrt{34}}{3}}$$

$$\textcircled{16} 5x^2 - 40x - 1 = 0 \quad \textcircled{17} -6x^2 - 12x + 19 = 0$$

$$\frac{5x^2 - 40x - 1 = 0}{5} \quad \frac{-6x^2 - 12x + 19 = 0}{-6}$$

$$x^2 - 8x - \frac{1}{5} = 0$$

$$x^2 - 8x = \frac{1}{5}$$

$$x^2 - 8x + 16 = \frac{1}{5} + 16$$

$$(x-4)(x-4) = \frac{1}{5} + \frac{16}{1}$$

$$(x-4)^2 = \frac{1}{5} + \frac{80}{5}$$

$$(x-4)^2 = \frac{81}{5}$$

$$\sqrt{(x-4)^2} = \pm \sqrt{\frac{81}{5}}$$

$$x-4 = \pm \frac{\sqrt{81}}{\sqrt{5}}$$

$$x-4 = \pm \frac{9}{\sqrt{5}}$$

$$x-4 = \pm \frac{9 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$x-4 = \pm \frac{9\sqrt{5}}{5}$$

$$\boxed{x = 4 \pm \frac{9\sqrt{5}}{5}}$$

$$\frac{-6x^2 - 12x + 19 = 0}{-6}$$

$$x^2 + 2x - \frac{19}{6} = 0$$

$$x^2 + 2x = \frac{19}{6}$$

$$x^2 + 2x + 1 = \frac{19}{6} + 1$$

$$(x+1)(x+1) = \frac{19}{6} + \frac{1}{1}$$

$$(x+1)^2 = \frac{19}{6} + \frac{6}{6}$$

$$(x+1)^2 = \frac{25}{6}$$

$$\sqrt{(x+1)^2} = \pm \sqrt{\frac{25}{6}}$$

$$x+1 = \pm \frac{\sqrt{25}}{\sqrt{6}}$$

$$x+1 = \pm \frac{5}{\sqrt{6}}$$

$$x+1 = \pm \frac{5 \cdot \sqrt{6}}{\sqrt{6} \cdot \sqrt{6}}$$

$$x+1 = \pm \frac{5\sqrt{6}}{6}$$

$$\boxed{x = -1 \pm \frac{5\sqrt{6}}{6}}$$

$$\textcircled{18} 3x^2 - 21x + 8 = 0$$

$$\frac{3x^2 - 21x + 8 = 0}{3}$$

$$x^2 - 7x + \frac{8}{3} = 0$$

$$x^2 - 7x = -\frac{8}{3}$$

$$x^2 - 7x + \frac{49}{4} = -\frac{8}{3} + \frac{49}{4}$$

$$(x - \frac{7}{2})(x - \frac{7}{2}) = \frac{-32}{12} + \frac{147}{12}$$

$$(x - \frac{7}{2})^2 = \frac{115}{12}$$

$$\sqrt{(x - \frac{7}{2})^2} = \pm \sqrt{\frac{115}{12}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{115}}{\sqrt{12}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{115}}{\sqrt{4} \cdot \sqrt{3}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{115}}{2\sqrt{3}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{115} \cdot \sqrt{3}}{2\sqrt{3} \cdot \sqrt{3}}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{345}}{2 \cdot 3}$$

$$x - \frac{7}{2} = \pm \frac{\sqrt{345}}{6}$$

$$\boxed{x = \frac{7}{2} \pm \frac{\sqrt{345}}{6}}$$

$$\textcircled{19} 0 = 7x^2 + 35x - 10$$

$$\frac{0 = 7x^2 + 35x - 10}{7}$$

$$0 = x^2 + 5x - \frac{10}{7}$$

$$\frac{10}{7} = x^2 + 5x$$

$$\frac{25}{4} + \frac{10}{7} = x^2 + 5x + \frac{25}{4}$$

$$= (x + \frac{5}{2})(x + \frac{5}{2})$$

$$\frac{175}{28} + \frac{40}{28} = (x + \frac{5}{2})^2$$

$$\frac{215}{28} = (x + \frac{5}{2})^2$$

$$\pm \sqrt{\frac{215}{28}} = \sqrt{(x + \frac{5}{2})^2}$$

$$\pm \frac{\sqrt{215}}{\sqrt{28}} = x + \frac{5}{2}$$

$$\pm \frac{\sqrt{215}}{\sqrt{4} \cdot \sqrt{7}} = x + \frac{5}{2}$$

$$\pm \frac{\sqrt{215}}{2\sqrt{7}} = x + \frac{5}{2}$$

$$\pm \frac{\sqrt{215} \cdot \sqrt{7}}{2\sqrt{7} \cdot \sqrt{7}} = x + \frac{5}{2}$$

$$\pm \frac{\sqrt{1505}}{2 \cdot 7} = x + \frac{5}{2}$$

$$\boxed{-\frac{5}{2} \pm \frac{\sqrt{1505}}{14} = x}$$